

Nanoelectronics Moving Merely At A Nanoscale

Theodore Von Kármán, a Hungarian-American aerospace engineer, mathematician and physicist, once said, "Scientists discover the world that exists; engineers create the world that never was." Nanoelectronics is an exciting field that is forecast to influence nearly every facet of our life. It definitely has a phenomenal future, but are there enough current opportunities to make a career out of it? Let us find out



Engineers working at a nanoelectronics computation lab (Image courtesy: www.ee.iitb.ac.in)

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Involvement with the utilisation of nanotechnology in electronic components, nanoelectronics is perhaps one of the increasingly important areas for the years and decades to come. From designing medical devices to building better batteries, and from manipulating molecules to manufacturing, nanoelectronics is truly a fascinating field to have a career in, though it does not yet have a significant presence in India.

Nanoelectronics in India

A unique initiative for accelerating research and development in nanoelectronics in India was launched in August 2008 at the Centers of Excellence in Nanoelectronics (CEN) at

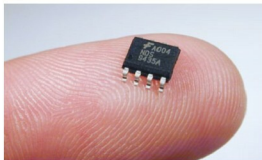
Indian Institute of Science, Bangalore (IISc) and Indian Institute of Technology, Bombay (IITB). The Indian Nanoelectronics Users Program (INUP) intends to facilitate experimentation of research ideas of Indian researchers in the general area of nanoelectronics. Conceived and funded by the Depart-

ment of Information Technology (DIT), the Ministry of Communications and Information Technology, government of India, this program aims at providing hands-on training to researchers from R&D organisations in India and facilitating advanced research by enabling implementation of their research work. Details about the INUP program can be found at <http://www.inup.org.in>.

The phenomenal growth in the CMOS technology over the past four decades has enabled very high performance compute and storage systems, powering the information technology revolution. The MOS transistor has metamorphosed from a few tens of microns dimension in 1970s to a few tens of nanometres today. Semiconductor memory and high-performance logic circuits have been the technology drivers to architect the miniaturisation of the MOS transistor, and this will continue in the foreseeable future. The underlying premise of this growth is the ability to miniaturise the MOS transistor dimension by 30 per cent,

in every two-year cycle on an average.

The scaling of MOS transistor in nanoelectronics era necessitates the exploration of new materials (high-k gate dielectrics such as HfO_2 , Er_2O_3 , Gd_2O_3 ; new channel materials such as germanium and graphene) and new device structures (double-gate FET, FinFET, Schottky source/drain FET). The facilities avail-



Nanoelectronics: Application of nanotechnology in the field of electronics and electronic components (Image courtesy: <http://childfriendlynews.com/tech-stuff-that-kids-should-know/>)